





SCALING MADE SIMPLE: HOW KUBERNETES HANDLES GROWING APPS

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ABOUT ME

Alvaro Revuelta.

- Systems Developer
- Laboratory for Life Sciences - SciLifeLab
- Follow the presentation



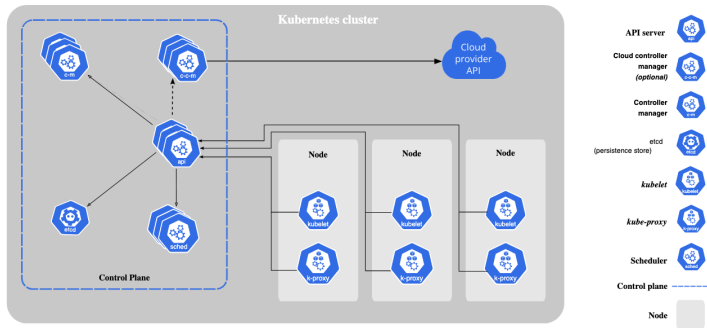


PRESENTATION STRUCTURE

1. K8s intro.
2. What is Scaling?
3. How Kubernetes Solves Scaling (Including live demo).
4. Final



1. K8S INTRO

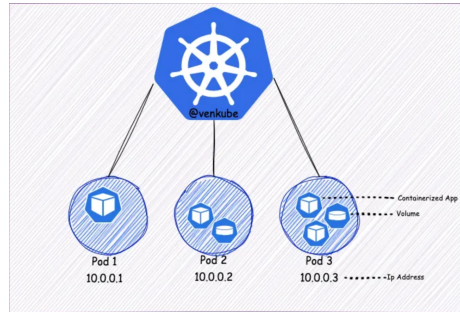


SOURCE: KUBERNETES.IO



1. K8S INTRO

- Open-source platform widely used in cloud infrastructures
- Managing of containers efficiently
- Key components
 - **Master Nodes.**
 - **Worker Nodes.**
 - **Pods:** Smallest deployable unit
 - **Deployments:** Kubernetes resource where we specify the number and characteristics of our pods.
 - **Controllers:** Ensure the desired state of the applications.



Source: [VenKube](#)



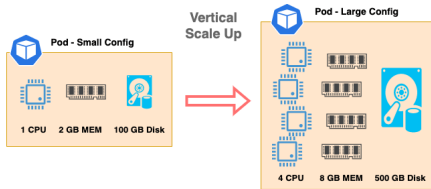
2. WHAT IS SCALING?

What happens when your app gets popular?

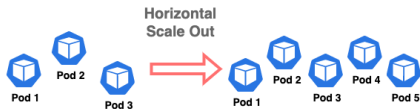
- **Scaling up** means adding more infrastructure, changing computing power or databases as needed.
- Ex: Larger hard drive, more CPU, etc.
- **Scaling out** is more associated with distributed architectures
- You would, for example, add more nodes to divide the workload among them.



3. HOW KUBERNETES SOLVES SCALING.



Source: [portworx](#)



Source: [portworx](#)



3. HOW KUBERNETES SOLVES SCALING

[Official Docs](#)

Scaling Out

- Manually adjust the number of pod replicas.
- Horizontal Pod Autoscaler (HPA) scales pods according to usage.

Scaling Up

- Manually adjust pod resources. (New pods created before the old ones are removed).
- Vertical Pod Autoscaler (not enabled by default.)
- [AWS Guide](#)



3. HOW KUBERNETES SOLVES SCALING.

DEMO TIME!



3. HOW KUBERNETES SOLVES SCALING.

Other interesting resources/concepts.

1. Cluster Autoscaler: Also not enabled by default.
2. Autoscaling based on events:
 - For example, based on the number of queries or messages in a queue.
 - Can be combined with an autoscaling based on time, specifying to add resources during peak hours.
 - Check [KEDA project](#).



4. FINAL

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4. FINAL

Q & A